

McGill University
Department of Mining and Materials Engineering



MIME 455
Advanced Process Engineering

**Project 1: The Impact of Flow Modifiers on Fluid Dynamics and
Non-Metallic Inclusion Removal in Tundish: Base Case vs. Dam
and Turnflow Case**

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Abstract

The influence of flow modifiers—a dam and turnflow configuration—on fluid dynamics and non-metallic inclusion removal in a steelmaking tundish is investigated in this project. Computational fluid dynamics simulations are performed using ANSYS FLUENT to generate velocity streamlines, velocity contours, and turbulent kinetic energy contours for two tundish designs: a base case and a modified case incorporating a dam. Governing fluid mechanics equations, including the continuity and momentum transport equations, energy balance, Reynolds number, and the standard k- ϵ turbulence model, are employed to characterize the turbulent flow of molten steel. It is found that the dam and turnflow configuration significantly improves internal flow behavior by enhancing recirculation, increasing residence time, and reducing dead zones. As a result, the flotation and removal of non-metallic inclusions are promoted, leading to improved steel cleanliness. The findings are used to underscore the importance of strategic flow control in tundish design for achieving superior metallurgical quality in continuous casting operations.

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1. Introduction

In steelmaking, a tundish is a refractory-lined vessel used as an intermediary between the teeming ladle and the continuous casting mold. It is employed to maintain a controlled and continuous flow of molten steel into the mold, while also facilitating the removal of impurities and the homogenization of temperature prior to solidification [1]. Dams are refractory barriers placed across the tundish floor to control molten steel flow. By obstructing the direct flow path, dams create a turn in the flow, forcing the molten steel to change direction. This redirection increases residence time, promoting the flotation and removal of non-metallic inclusions and improving steel cleanliness [2]. Dams also aid in homogenizing temperature and composition before casting.

In this project, the fluid flow behavior within a tundish is investigated by comparing a base case with a modified design that incorporates a dam to create turn flow as a flow modifier. The simulation is carried out using the ANSYS FLUENT CFD Solver, where computational fluid dynamics techniques are applied to model the turbulent flow of molten steel within the tundish. The modeling framework incorporates the continuity equation, the momentum transport equation, the energy equation, and the standard k- ϵ turbulence model. The Reynolds number is also considered to characterize the flow regime and ensure appropriate turbulence modeling.

Additionally, the potential benefits of implementing pour pads—strategically positioned at the impact point of the incoming steel jet on the tundish floor—are evaluated. The objective of this project is to observe and analyze the flow patterns generated by the flow modifier and to assess its impact and effectiveness in enhancing the removal of non-metallic inclusions through improved fluid dynamics, ultimately leading to higher purity and better product quality.

2. Theory

This section outlines the fundamental equations and principles governing fluid flow and energy transport, including the continuity and momentum equations, the energy balance, Reynolds number, the k- ϵ turbulence model, and the boundary conditions, which form the basis for the simulations and analyses presented in this work.

2.1 Continuity Equation

In fluid dynamics, the continuity equation embodies the principle of mass conservation, stating that mass is neither created nor destroyed as fluid flows through space and time [3]. In its general form, the continuity equation is written as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{q}) = 0 \quad \text{Eq. 1}$$

where ρ is the fluid density and $\mathbf{q}$ is the velocity vector. This equation states that the time rate of change of density at a point plus the divergence of the mass flux must be zero. In Cartesian coordinates, this expands to:

$$\frac{\partial \rho}{\partial t} + \rho \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) = 0 \quad \text{Eq. 2}$$

where v_x , v_y and v_z are the velocity components in the x, y, and z directions. For incompressible flows, where the fluid density remains constant (fluid particles do not undergo significant changes in density as they move), the time derivative of density $\frac{\partial \rho}{\partial t}$ becomes zero and can be eliminated from the continuity equation. Equation 2 can be simplified to

$$\rho \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) = 0 \quad \text{Eq. 3}$$

For the flow within the tundish, Equation 3 ensures conservation of the volumetric flow rate. Since the density of the molten metal remains essentially constant during flow, it is considered incompressible, allowing the use of this simplified form of the continuity equation.

2.2 Momentum Transport Equation

The momentum transport equation, commonly known as the Navier–Stokes equation, governs the motion of fluids by applying Newton’s second law to a fluid element. It is written as:

$$\rho \left(\frac{\partial \vec{v}}{\partial t} \right) + \rho (\vec{v} \cdot \nabla) \vec{v} = -\nabla p + \mu \cdot \nabla^2 \vec{v} + \rho \vec{g} + F_\sigma \quad \text{Eq. 4}$$

In this equation, the left-hand side represents the inertial terms, which include the local acceleration $\rho \left(\frac{\partial \vec{v}}{\partial t} \right)$ and the convective acceleration $\rho (\vec{v} \cdot \nabla) \vec{v}$, accounting for how fluid velocity changes over time and space. The right-hand side consists of the forces acting on the fluid: the pressure gradient force $-\nabla p$, viscous forces $\mu \cdot \nabla^2 \vec{v}$, gravitational body force $\rho \vec{g}$, and any additional external or surface forces F_σ . The Momentum Transport Equation describes how momentum is transferred within a fluid and plays a crucial role in accurately modeling fluid behavior.

For the flow within the tundish, assuming steady flow conditions, the velocity at any given point does not change with time. Consequently, the local acceleration term $\rho \left(\frac{\partial \vec{v}}{\partial t} \right)$ is negligible. Additionally, since there are no additional external or surface forces, the term F_σ is also omitted. The Navier–Stokes equation reduces to:

$$\rho (\vec{v} \cdot \nabla) \vec{v} = -\nabla p + \mu \cdot \nabla^2 \vec{v} + \rho \vec{g} \quad \text{Eq. 5}$$

2.3 Energy Balance in Steady-Flow Systems

Following the mass and force balances, energy balances are considered using the Steady Flow Energy Equation under steady-state conditions, as shown in Equation 6. It represents an application of the 1st law of thermodynamics to a control volume with continuous fluid flow, ensuring the conservation of energy under steady-state conditions [4].

$$\dot{Q} - \dot{W}_s = \dot{m} \left[h_2 + \frac{1}{2}U^2 + gz_2 - h_1 + \frac{1}{2}U^2 + gz_1 \right] \quad Eq. 6 [4]$$

where $\dot{Q}$ represents the rate of heat transfer into the system, $\dot{W}_s$ represents the external shaft work associated with fluid motion and energy losses due to friction, $\dot{m}$ is the mass flow. The variables h , U , and z correspond to specific enthalpy, velocity, and elevation, respectively, with subscripts 1 and 2 referring to inlet and outlet conditions. In this study, the process is assumed to be isothermal—occurring at a constant temperature—thereby eliminating heat transfer. As no external rotating components (e.g., compressors) are present, shaft work $\dot{W}_s$ is considered to arise solely from viscous dissipation due to turbulence. This assumption is supported by the high Reynolds number calculated in Section 2.4, which is indicative of fully developed turbulent flow. Equation 6 can be simplified as shown below, highlighting the role and effects of viscous dissipation on changes in the fluid's enthalpy, kinetic energy, and potential energy.

$$- \dot{W}_s = \dot{m} \left[h_2 + \frac{1}{2}U^2 + gz_2 - h_1 + \frac{1}{2}U^2 + gz_1 \right] \quad Eq. 7$$

2.4 Reynolds Number

The Reynolds Number (Re), a dimensionless quantity, is used to predict flow patterns in various fluid flow scenarios and is defined as the ratio of inertial forces to viscous forces [5]. The Reynolds number is calculated below using Equation 8.

$$Re = \frac{\rho UL}{\mu} \quad Eq. 8$$

When the calculated Reynolds number exceeds 2300, it typically indicates the presence of turbulent flow. However, the onset and nature of turbulence also depend on the geometry of the system. In the case of the tundish, the region near the inlet—often referred to as the active zone, includes the incoming flow interacts with the walls—exhibits turbulent behavior due to high velocity and mixing. In contrast, regions further from the inlet, known as dead zones, experience relatively steady and less dynamic flow. This illustrates the coexistence of both turbulent and steady flow within the tundish: while localized areas show turbulent motion, the overall system maintains steady-state conditions in a time-averaged sense.

2.5 K-ε turbulence model

The Reynolds number calculated exceeds the critical threshold for turbulence, justifying the use of a k-ε turbulence model in the simulation. The k-ε model was chosen over k-ω due to its suitability for high Reynolds number flows, faster performance, and greater numerical stability, making it ideal for this project. The governing equations include the turbulence kinetic energy k , as shown in Equation 9, and turbulence kinetic energy dissipation ε , presented in Equation 10.

$$\frac{\partial(\rho k)}{\partial t} + \nabla \cdot (\rho \vec{v} k) = \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right] + G_k - \rho \varepsilon \quad Eq. 9$$

On the left-hand side, the term $\frac{\partial(\rho k)}{\partial t}$ is used to represent the unsteady rate of change of turbulent kinetic energy accumulation, while $\nabla \cdot (\rho \vec{v} k)$ accounts for its convection by the mean flow. On the right-hand side, diffusion is modeled through the term $\nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right]$ where contributions from both molecular viscosity μ and turbulent viscosity μ_t are considered and scaled by the turbulent Prandtl number σ_k . Production of turbulent kinetic energy is captured by the term G_k , and dissipation is accounted for by $\rho \varepsilon$, representing the rate at which turbulent energy is converted into thermal energy.

$$\frac{\partial(\rho \varepsilon)}{\partial t} + \nabla \cdot (\rho \vec{v} \varepsilon) = \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \nabla \varepsilon \right] + C_1 \frac{\varepsilon}{k} G_k - C_2 \frac{\varepsilon^2}{k} \rho \quad Eq. 10$$

For equation 10, the rate of change of ε over time is captured by the term $\frac{\partial(\rho \varepsilon)}{\partial t}$, while $\nabla \cdot (\rho \vec{v} \varepsilon)$ accounts for its convection by the mean velocity field. The diffusion of ε is modeled through the term $\nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \nabla \varepsilon \right]$, where molecular viscosity μ and turbulent viscosity μ_t are considered, scaled by the turbulent Prandtl number for ε , σ_ε . The production of dissipation is represented by $C_1 \frac{\varepsilon}{k} G_k$, which links the generation of turbulence to the energy dissipation rate, with c_1 being an empirical constant. Final term $C_2 \frac{\varepsilon^2}{k} \rho$ accounts for the natural decay or destruction of ε , with C_2 also being a model constant.

For this project, all solid surfaces of the tundish are subjected to non-slip boundary conditions, and the simulation is performed under steady-state conditions. Therefore, the gradients of turbulent kinetic energy and its dissipation rate are zero at both the symmetry planes and the free surface of the bath. Accordingly, Eq 9 and 10 can be simplified as follows:

$$\nabla \cdot (\rho \vec{v} k) = \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right] + G_k - \rho \varepsilon \quad Eq. 11$$

$$\nabla \cdot (\rho \vec{v} \varepsilon) = \nabla \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \nabla \varepsilon \right] + C_1 \frac{\varepsilon}{k} G_k - C_2 \frac{\varepsilon^2}{k} \rho \quad \text{Eq. 12}$$

The standard k-ε turbulence model is employed in this project to simulate both the tundish-based case and the dam case. The specific constants used for the k-ε model are listed in Table 1 below.

Table 1. Values of the Constants in the Standard k-ε Model

Constant	Values
C_μ	0.09
C_1	1.44
C_2	1.92
σ_k	1.0
σ_ε	1.3

2.6 Boundary Conditions, Parameters, and Hypotheses

The boundary conditions for the ANSYS simulation in Project 1 are carefully defined to capture how each section of the process design influences the fluid dynamics of molten steel within the tundish. These conditions are essential for ensuring accurate system modeling and for enabling the estimation and comparison of various flow behaviors of liquid steel—both in the base case and in the cases with flow modifiers (dam and turnflow). The Project 1 boundary conditions for the computational fluid dynamics simulation—applicable to both the base case and the modified configurations—are defined as follows:

- **Inlet:** The inlet is defined at the pour-pad entry where liquid steel enters the domain with an initial velocity of 0.8 m/s. This velocity condition dictates the flow behavior throughout the simulation domain.
- **Outlet:** The outlet is modeled as a pressure outlet with a gauge pressure of 0 Pa to simulate atmospheric conditions. Default turbulence settings are used to maintain solution stability in the event of reverse flow.
- **Slag:** The slag layer is represented as the upper boundary of the molten steel bath. It is modeled with a zero-shear condition, indicating the absence of tangential stress between the slag and molten steel.

- Walls: All tundish walls are treated as stationary with a no-slip boundary condition, implying a fluid velocity of 0 m/s at the wall surface. The wall material is assumed to be aluminum, as specified by FLUENT's default settings.
- Symmetry Plane: A symmetry boundary condition is applied to exploit the geometric symmetry of the tundish. This reduces the computational domain and enhances visualization efficiency for the fluid flow behavior.
- Dam (flow modifiers case): A 25 mm thick, 200 mm high dam is placed 500 mm from the outlet to visualize the influence of flow modifiers on fluid flow and inclusion removal. The dam is assigned a wall boundary condition.
- Turnflow (flow modifiers case): A turnflow device with a diameter of 125 mm and a thickness of 15 mm is placed and aligned with the inlet to further analyze the influence of flow modifiers on the fluid flow and inclusion removal. The turnflow is assigned a wall boundary condition.

The simulation is conducted under isothermal conditions, meaning the temperature is maintained uniformly and remains constant throughout the entire process. Other assumptions include steady-state conditions, a single-phase system (steel) with no interaction with other materials, and the use of the standard k- ϵ turbulence model. Once all boundary conditions are defined, the simulation can generate accurate computational fluid dynamics simulation results to assess the impact of flow modifiers on tundish design and metallurgical processing. Figures 1 show the base case and modified geometries, respectively, with labeled boundaries.

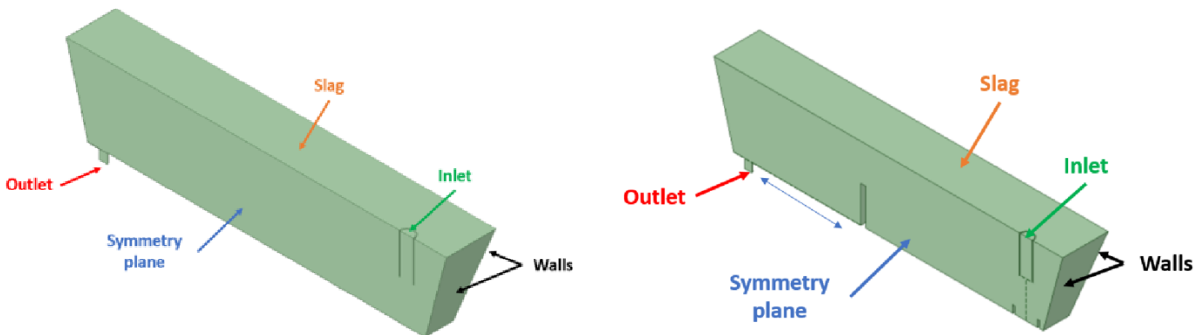


Figure 1. Tundish Base Case and Modified (Dam and Turnflow) Cases Geometries

3. Computational Method

Two simulation cases are conducted for project 1 to analyze the impact of flow modifiers on fluid flow, aiming to enhance the removal of non-metallic inclusions. The simulations are carried out using ANSYS Fluent 2025 R1. This section introduces the computational methodology used in the simulations, with a focus on optimizing both accuracy and computational efficiency throughout the computational fluid dynamics analysis.

3.1 Meshing

Once the geometry was finalized, meshing was performed using the Workbench Meshing module. In this step, the model was broken down into finite elements, allowing fluid flow to be numerically analyzed by ANSYS. A finer mesh was used to enhance resolution and accuracy, leading to more reliable predictions, especially when the influence of flow modifiers on fluid behavior was assessed.

A uniform element size of 0.02 m was applied, and a target skewness of 0.6 was maintained to ensure mesh quality. The sizing method was applied using face sizing, and all model faces were selected—12 for the base case and 20 for the flow modifier case. The mesh was generated with consistent element sizing and controlled skewness throughout.

The final mesh was composed of 114,338 elements for the base case and 114,666 elements for the flow modifier case. The average skewness was recorded at 0.2758 for the base case and 0.2768 for the modifier case, indicating high mesh quality in both cases. The slight increase in skewness for the modifier case was expected due to the added geometric complexity, but the values remained well below the acceptable threshold of 0.6, confirming that high-quality meshing was achieved across both scenarios.

3.2 General Setup:

A pressure-based solver was used with an absolute velocity formulation under steady-state conditions. The k-epsilon turbulence model was applied, with constant values assigned according to the project guidelines. Gravity was enabled in the Z-direction with a value of -9.81 m/s^2 to reflect realistic fluid flow conditions within the tundish.

3.3 Solver Configuration and Solution Initialization

A coupled pressure-velocity scheme was selected as the primary solver configuration due to its well-established convergence stability and its effectiveness in accurately resolving pressure-velocity interactions. Default settings were maintained for the discretization schemes, with a first-order upwind method applied to both the turbulent kinetic energy and turbulent dissipation rate. This configuration was adopted to provide a balance between numerical stability and computational efficiency during the initial stages of the simulation. In accordance with the project manual, residuals for k and ϵ were set to 0.01 to ensure satisfactory convergence of the simulation results.

3.4 Iterative Process and Simulation of Streamlines and Contours

The simulation was configured to run for a maximum of 1000 iterations or until the convergence criteria were met. Through this iterative approach, progression toward a fully resolved and stable flow field was ensured, representative of a true steady-state condition. An iteration limit of 1000

was provided to allow the system to be thoroughly analyzed by ANSYS, and to enable verification of convergence for the accuracy and correctness of the simulation to be confirmed.

All simulation data were stored within the Solution module of ANSYS Fluent. Velocity streamlines for both the base and modified cases were generated within a uniform range of 0 to 0.25 m/s to allow for direct visual comparison. Similarly, contours of turbulent kinetic energy were produced within a range of 0 to 0.05 m²/s², and velocity magnitude contours were visualized from 0 to 0.8 m/s. Consistent value ranges were applied across both cases to ensure that differences in flow behavior were not influenced by scale variations, thereby improving the accuracy of result interpretation. Overall, distinct fluid flow patterns were observed in each case and will be analyzed in Section 4: Results and Discussion, to study the role of flow modifiers in enhancing the removal of non-metallic inclusions during material processing.

4. Result and Discussion

After performing different simulations, the results of the tundish base case and the dam plus turnflow modified case are presented and compared. In this section, the velocity streamlines, velocity contours, and turbulent kinetic energy contours for both cases are analyzed, and the effects of including the dam and turnflow configuration are discussed.

4.1 Velocity Streamlines

Figure 2 illustrates the base-case velocity streamlines within the tundish. A high-velocity region (shown in red) is formed near the inlet, with velocities reaching approximately 0.25 m/s. A concentrated jet is subsequently created, which is dispersed and curved upward due to interactions with the surrounding fluid. As the flow progresses toward the center of the domain, it is observed to transition through shades of green and blue—indicating that the flow becomes smoother, decelerated, and more diffusive. Just to the right of the inlet, a noticeable vortex is formed as a result of the jet's proximity to the vertical wall. The boundary layer is disturbed by the high-speed jet, inducing rotational motion and forming a localized recirculation zone, by which mixing is enhanced.

Overall, the high velocities are observed near the inlet due to the gravitational force of the incoming stream, while a gradual decrease in velocity is seen toward the center and left side of the tundish. Near the outlet, velocity is observed to increase again, forming a secondary high-velocity region. Although this high velocity is considered beneficial for the removal of non-metallic inclusions with the outgoing steel, careful control is required to prevent the re-entrainment of inclusions due to turbulence. These shifts in velocity are attributed to the combined effects of gravitational forces and momentum transfer, and mark the transition from active to dead zones.

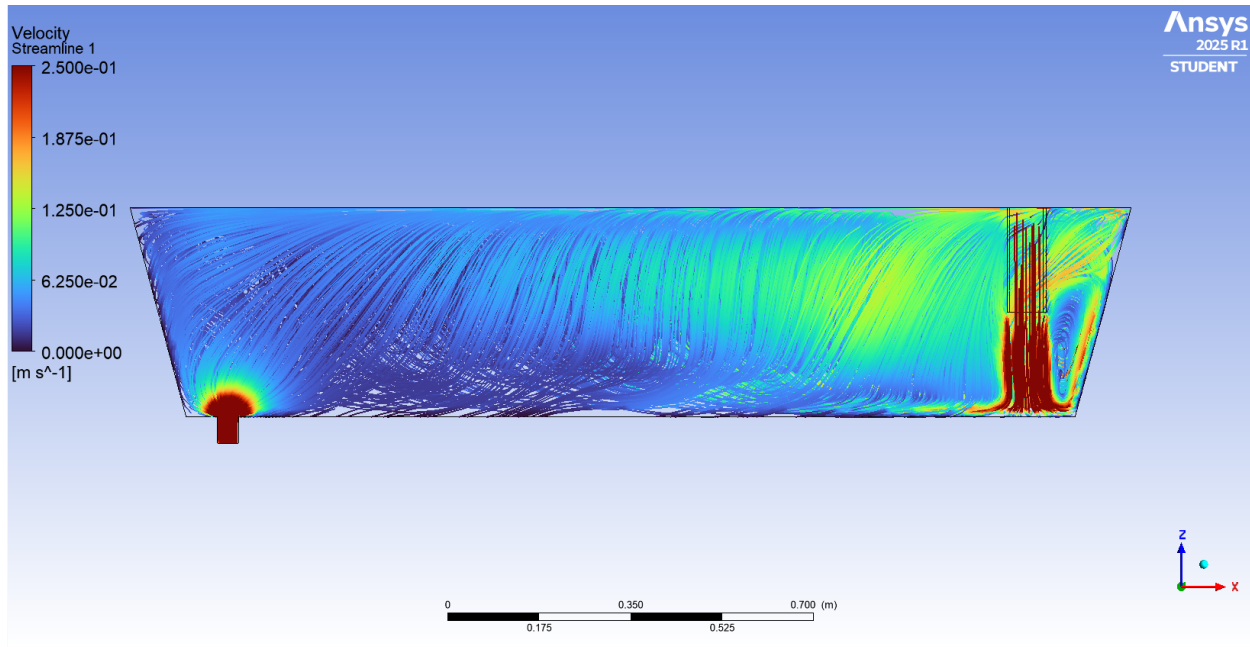


Figure 2. Velocity Streamlines – Base Case

The streamline distribution for the modified case, featuring a dam and turnflow structure, is shown in Figure 3, while the corresponding isometric view is depicted in Figure 4. As in the base case, a high-velocity jet is introduced from the inlet, indicated in red region at 0.25 m/s. Just beneath the inlet, the incoming fluid is redirected by a turnflow feature into two high velocity up-moving jets, resulting in strong initial momentum and enhanced circulation near the inlet zone. Interaction with the surrounding fluid occurs quickly, and a broad recirculation region is formed. The symmetric red flow pattern observed at the inlet is primarily attributed to the alignment between the inlet geometry center and the turnflow center, along with the influence of gravitational force.

At the center of the domain, the flow path is disrupted by the dam structure, and the domain is divided into upstream and downstream vortex regions. As the flow approaches the dam, it slows down, as indicated by the dark blue streamlines. This deceleration reveals the formation of a large downstream vortex, caused by fluid accumulation. The momentum on the downstream side is regained as the redirected flow is guided in an arcing path toward the outlet. At the outlet, the fluid is accelerated again, as observed in the base case.

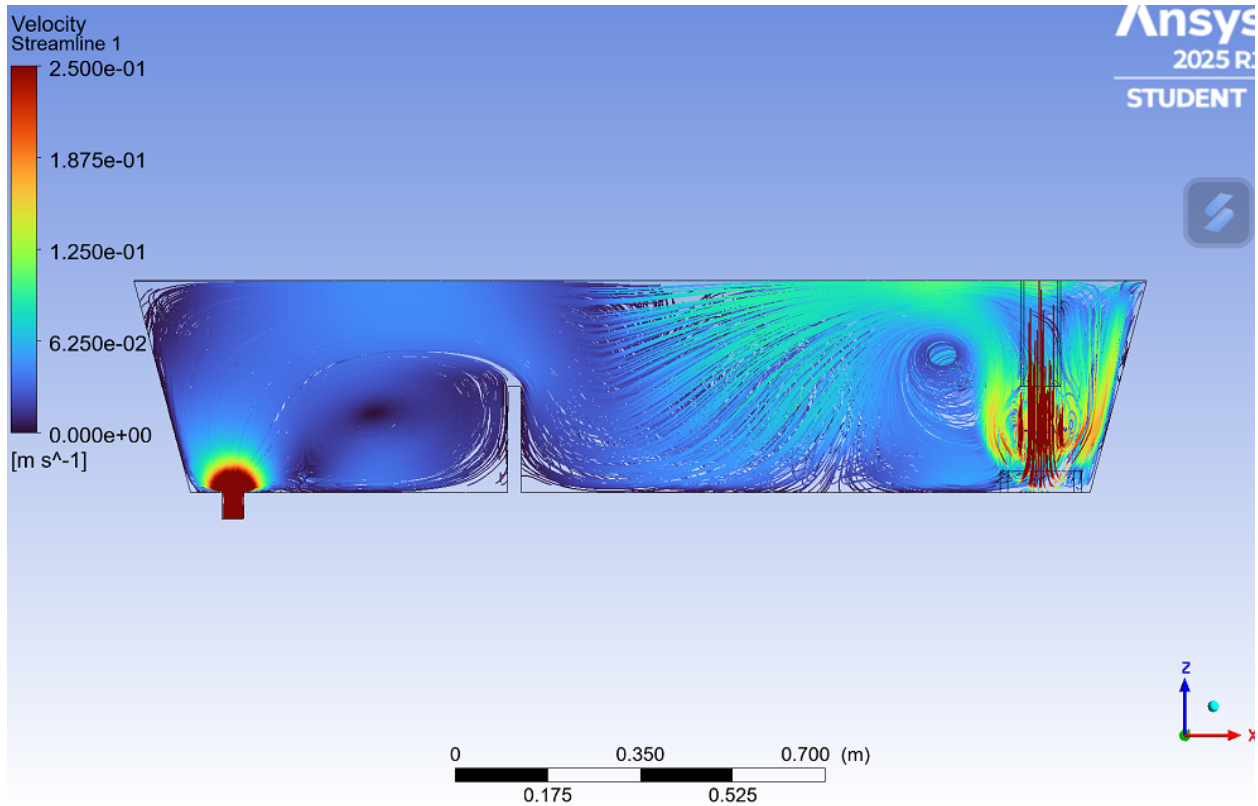


Figure 3. Velocity Streamlines – Dam and Turn Flow Case

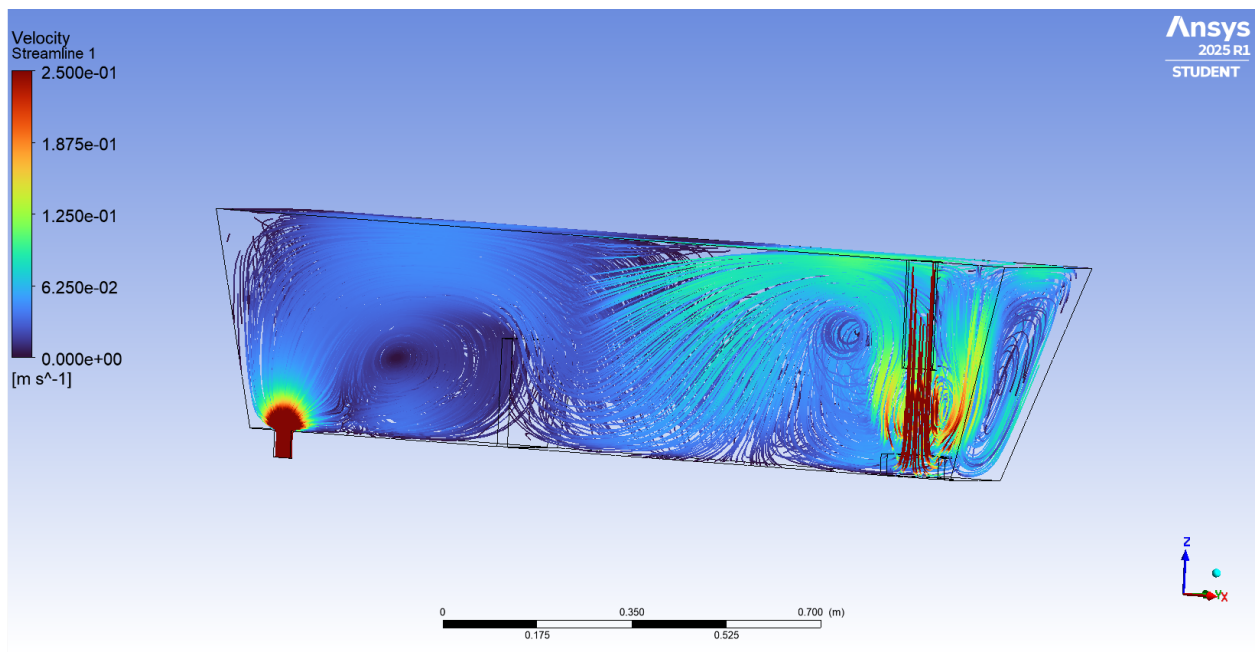


Figure 4. Isometric View of Velocity Streamlines – Dam and Turnflow Case

The comparison of velocity streamlines clearly demonstrates the beneficial impact of the dam and turnflow on the internal fluid dynamics of the tundish. In the base case, the tundish floor is struck

directly by the incoming jet due to the absence of dams, resulting in high local velocities that disturb settled inclusions. The lack of strong recirculation zones shortens residence time, limiting the removal of inclusions.

The addition of a dam and turnflow creates multiple recirculation loops, enhancing residence time and promoting effective flotation and removal of inclusions. This recirculation improves mixing and directs some upward-moving flow toward the surface, encouraging inclusions and impurities to rise and separate from the molten steel. Additionally, dead zone volume is reduced, keeping the molten steel in motion and minimizing dead zones where inclusions might become trapped. Overall, a more favorable environment for non-metallic inclusion removal is established, resulting in a cleaner steel product.

4.2 Velocity Contour

The velocity contour is used to provide a clear visualization of the velocity variation across different regions within the tundish. In the base case shown in Figure 5, a strong jet of liquid steel enters through the inlet and strikes the tundish floor, creating a high-velocity zone near the inlet, shown in light red. The flow reaches a maximum velocity of approximately 0.87 m/s, with this peak concentrated around the outlet regions, as indicated by the dark red areas. The higher velocity at the outlet compared to the inlet is primarily due to its narrower diameter (40 mm for outlet versus 65 mm at the inlet). According to the continuity equation ($A_1 V_1 = A_2 V_2$), a smaller cross-sectional area at the outlet results in an increased flow velocity. Additionally, part of the potential energy from the tundish height is converted into kinetic energy, further contributing to the increased outlet velocity. The majority of the tundish volume is observed to fall within a lower velocity range of 0.09 to 0.18 m/s, particularly in the central part.

A relatively uniform and continuous distribution of flow velocity is also demonstrated by the base case velocity contour. Each velocity band, represented by consistent color gradients—from darker to lighter shades of blue—is seen occupying well-defined regions without frequent or abrupt transitions between different velocity levels. This suggests that the same flow velocity is maintained over broad areas, rather than being subjected to constant mixing or fluctuation at small spatial scales. As a result, relatively stable flow behavior is revealed, along with the presence of large dead zones, indicating limited mixing and low turbulence within the bulk of the tundish.

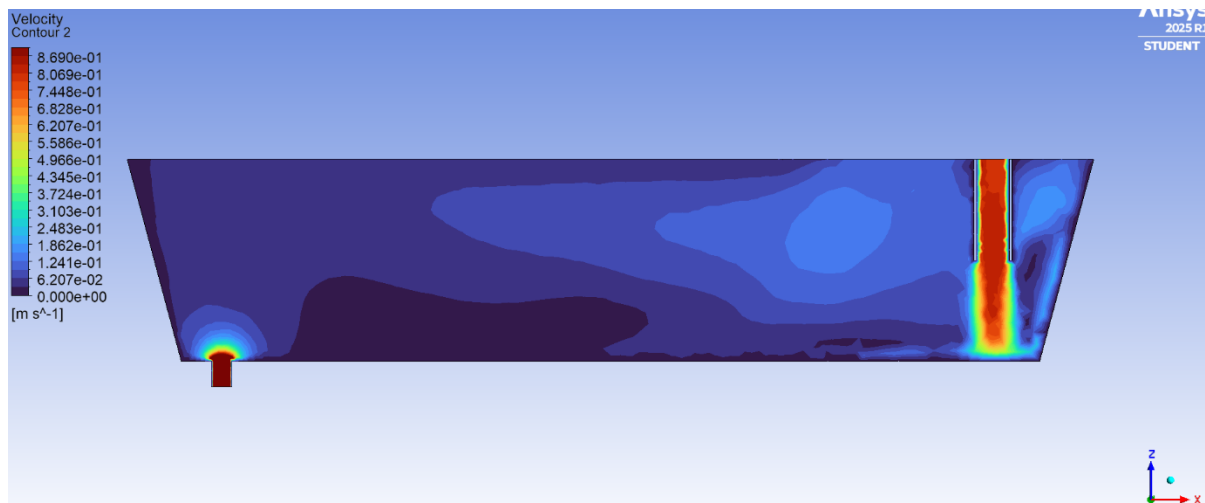


Figure 5. Velocity Contour – Base Case

The velocity contour of the Dam and Turnflow case, shown in Figure 6, reveals a strong inlet jet entering from the left, similar to the base case. The velocity remains concentrated at both the inlet and outlet regions, with a higher velocity observed at the outlet due to the same reason mentioned in the base case. Due to the influence of the turning flow, the inlet stream splits into two upward-moving streams with relatively high velocity, resulting in a larger active zone near the inlet compared to the base case.

In the dam and turnflow case, the velocity bands are observed to be more fragmented and dispersed, indicating a higher degree of mixing and more frequent variations in flow velocity. The swirling motion upstream (near the inlet) and downstream (near the dam) can still be partially identified through the velocity contour. Within these swirling regions, relatively low velocities (dark blue) are observed, confirming that vortex formation induced by the dam and turnflow has extended the residence time and promoted more effective flotation and inclusion removal.



Figure 6. Velocity Contour – Dam and Turn Flow Case

The velocity contour comparison highlights the influence of flow modifiers in enhancing flow distribution—particularly the development of swirling regions—and reducing stagnant areas (dead zones). In the dam configuration, a longer residence time within swirling regions is evident, along with a more active inlet zone caused by the turnflow. Additionally, frequent variations in velocity gradients throughout the tundish volume suggest that higher turbulence and improved flow engagement are achieved. Consequently, the extent of dead zones is significantly reduced in the dam and turnflow case, resulting in a more dynamic flow environment that is favorable for inclusion flotation and improved metallurgical cleanliness.

4.3 Turbulent Kinetic Energy Contour

The turbulent kinetic energy (TKE) contour is a key indicator of turbulence intensity within the tundish, helping identify areas with active fluid mixing. In the base case, shown in Figure 7, the highest TKE values—approximately $5.0 \times 10^{-2} \text{ m}^2/\text{s}^2$ —are observed near the outlet where the fluid is discharged. Relatively high TKE values, ranging from about $6 \times 10^{-3} \text{ m}^2/\text{s}^2$ to $1.6 \times 10^{-2} \text{ m}^2/\text{s}^2$, are also recorded at the bottom part of the inlet. TKE localized in compact zones and rapidly diminishing throughout the rest of the tundish volume. The central region, as well as much of the upper and bottom corners, exhibit very low TKE values, remaining below $2.6 \times 10^{-3} \text{ m}^2/\text{s}^2$, which points to a lack of significant turbulent motion. The contour reveals a stable flow field with large stagnant zones (dead zones), offering limited mixing and unfavorable conditions for inclusion flotation.



Figure 7. Turbulent Kinetic Energy Contour – Base Case

In the dam and turnflow case shown in Figure 8, turbulence is observed to be most prominent near the inlet and outlet. The inlet region is characterized by broader, more symmetrical areas with

elevated TKE, primarily attributed to the influence of the turnflow. The central body of the tundish continues to display uniform, low turbulence levels.

It is worth noting that, in the dam case, two swirling regions are visible in the velocity contour and streamline plots. Theoretically, these should correspond to higher TKE values and shown in Figure 8. However, the $k-\varepsilon$ turbulence model used in this project is relatively basic and may not fully capture such localized increases. These elevated TKE regions would likely be more apparent with a more advanced turbulence model. The outlet region continues to exhibit the highest TKE.

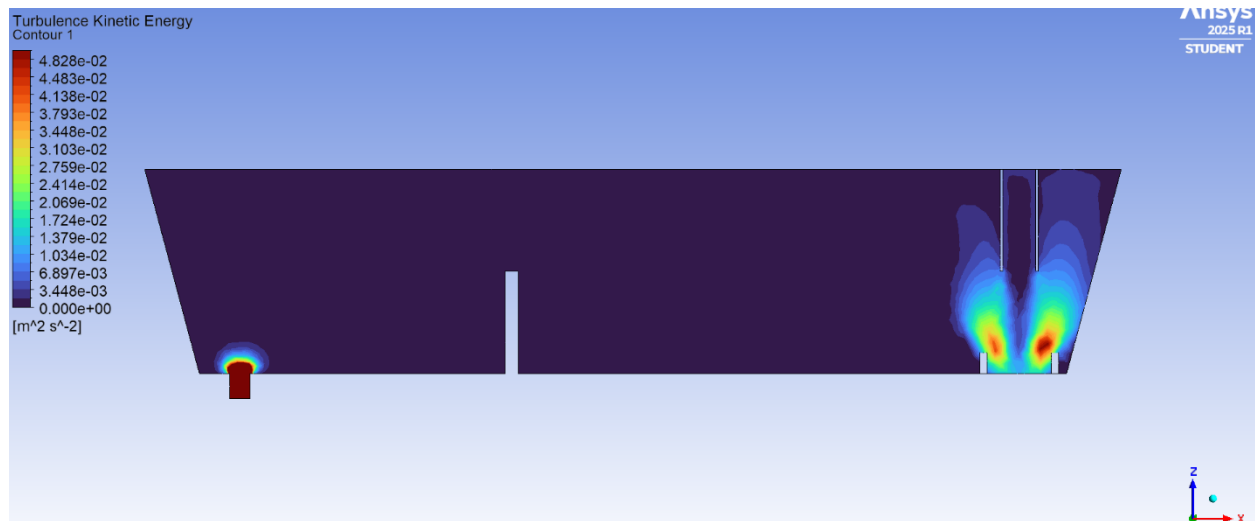


Figure 8. Turbulent Kinetic Energy Contour – Dam and Turn Flow Case

The main difference between the base case and the dam case is observed at the inlet. In the dam case, broader and more active inlet turbulence is produced due to the presence of turnflow. Through this mechanism, the inlet's kinetic energy is effectively converted into controlled mixing, as the turnflow transforms the high-speed jet into turbulent motion that engages the surrounding fluid. The flow's energy is intentionally utilized to stir the tundish contents, rather than being allowed to form narrow, direct jets that bypass the bulk volume. As a result, energy is not only dissipated more efficiently, but the internal flow field is also shaped in a manner that better supports metallurgical objectives.

5. Conclusion

The critical role of flow modifiers in optimizing tundish performance, particularly in enhancing non-metallic inclusion removal, is underscored by this project. Through computational analysis conducted in ANSYS, the base case is compared with a configuration incorporating a dam and turnflow. It is revealed that internal fluid dynamics—and, consequently, steel quality—are profoundly influenced by strategic flow control.

In the Base Case, where no flow modifiers are used, direct, high-velocity streams are observed, leading to turbulence near the inlet and outlet. These flow characteristics result in limited mixing zones and extensive dead volumes, which are known to hinder inclusion flotation and removal. In the modified tundish, a more purposeful flow pattern is introduced. The molten steel is redistributed by the dam, flow residence time is extended, upward motion is promoted, and dead zone is reduced—conditions essential for the flotation and removal of inclusions at the slag-molten steel interface. The turnflow enhances these effects by guiding the incoming jet upward and establishing structured recirculation zones.

As a result of these modifications, a more balanced flow environment is achieved, and turbulence is concentrated in beneficial zones. The possibility of inclusion re-entrainment is reduced, and dead zone that typically impede clean steel production are minimized. Overall, the integration of a dam and turnflow improves the tundish's capacity for controlled fluid motion and effective inclusion separation. Dam and turnflow provide valuable insights for tundish design optimization and contribute directly to the advancement of cleaner steelmaking processes.

References

- [1] “The Traveling Reservoir: The role of Steelmaking Tundish,” May 15, 2024.
<https://blog.cabaro-group.com/the-traveling-reservoir-the-role-of-steelmaking-tundish/>
- [2] “Fluid flow, flow control and modifying devices in a Tundish – IspatGuru.”
<https://www.ispatguru.com/fluid-flow-flow-control-and-modifying-devices-in-a-tundish/>
- [3] “Continuity equation.” https://www.princeton.edu/~asmits/Bicycle_web/continuity.html
- [4] D. E. Winterbone and A. Turan, “Introduction and revision,” in Elsevier eBooks, 2015, pp. 1–12. doi: 10.1016/b978-0-444-63373-6.00001-0.
- [5] N. Ansari and N. Ansari, “What is Reynolds Number?,” Ansys, Mar. 25, 2025.
<https://www.ansys.com/blog/what-is-reynolds-number>